Two-Transition Hybrid-Core Bayesian Inference: Code Package Summary

Summary of the code package built from `neutronstar_solver.py`, with validation results and caveats. Tested end to end on `EOSBetaFSU2L80.dat` with synthetic data only.

Status

The code package is written and has run end to end on the test EOS, but only against synthetic data, so there are no real posteriors yet. The NICER and GW170817 sample files were not available in the chat, so `config.DATA` has placeholder paths for them.

What's in the package

The whole pipeline sits in one folder (`/mnt/user-data/outputs/hybrid_eos_inference/`): `config.py`, `hybrid_eos.py`, `solver.py`, `priors.py`, `likelihood.py`, `model.py`, `run_inference.py`, `analysis.py`, `validate.py`, `mockdata.py` and a README.

- **Editing priors:** every prior range, tolerance, data path, branch rule and sampler setting is in `config.py`.
- **Hybrid EOS:** the model is hadronic table, jump, polytrope 1, jump, polytrope 2. Here ε_1 is the pre-jump top of polytrope 1 and P_t = P_h(ε_trans). The polytrope constants follow from pressure continuity:

```
K_1=\frac{P_t}
{(\varepsilon_\text{trans}+\Delta\varepsilon_\text{trans})^{\gamma_1}},\qquad
K_2=\frac{P_1}{(\varepsilon_1+\Delta\varepsilon_1)^{\gamma_2}}
```

- **Solver, rewritten from the original:** same equations and variables, now compiled with numba.
  - It integrates piecewise and stops exactly at each transition pressure.
  - It applies the corrected Takátsy–Kovács jump condition (in the solver's dimensionless variables):

```
\Delta y=-\frac{\bar R^3\,\Delta\bar\varepsilon}{\bar M+\bar R^3\bar P}
```

- It keeps all stable branches by the turning-point rule, so third-family branches are not dropped.
- **Likelihood:** NICER (M, R) KDEs marginalized over mass, a (Λ1, Λ2) KDE for GW170817, and the J0952 mass floor.

Run commands

- `python validate.py` runs the checks below.
- `python run_inference.py --nprocs 8` runs `dynesty` once per hadronic EOS and reports ln B against the pure-hadronic baseline.
- Add `--rule max` or `--rule sum` for the other twin-branch rules.
- `python analysis.py results/<name>` summarizes a finished run.

Validation results

- **Against the original equations (pure hadronic):** the mass agrees to about 1e-7, Λ to about 2e-4 or better, and the radius to about 1e-3 or better. The radius difference comes from where the crust surface is defined.
- **Jump condition:** integrating through a very thin smoothed transition converges to the corrected formula ($y_R \approx 1.033$) and not to the original Postnikov form (1.0215). For that EOS the original form shifts Λ by 0.9%.
- **Convergence:** at the default tolerance 1e-6 with 70 pressures, M_{TOV} is off by about 1e-7, $R_{1.4}$ by about 1e-3 km and $\Lambda_{1.4}$ by about 1e-5 relative to a much finer reference. The original solver used 1e-8 and 3000 pressures.
- **Love number:** the double-precision k2 matches mpmath to better than 1e-6 for $\beta \geqslant 0.005$.
- **Seidov criterion as a cross-check:**
  - It agrees with the computed stability in 95 of 107 prior draws.
  - All 12 disagreements are ill-posed cases (the transition sits at the hadronic mass maximum, or within about 0.01 of the threshold).
  - Only 4 draws pass the strict well-posed filter, and all 4 agree, so this check is weak evidence.
- **End-to-end sampler run:** a smoke test with `dynesty` on synthetic data gave ln B ≈ −1.7 against the hadronic truth, i.e. the hybrid is mildly disfavored when the data come from a hadronic EOS.

Things to know before running

- **Speed:** one likelihood call takes about 0.07 s on a single core, so use `--nprocs` for real runs.
- **Prior rejections:** about 63% of prior draws are rejected for causality. That is mostly the γ upper limit of 10, so narrowing the γ ranges in `config.py` would save a lot of compute.
- **Twin branches:** about 3% of accepted prior draws have more than one stable branch (3 of 111). On those 3 synthetic-data cases, switching branch rule moved ln L by about 0.1–0.2. A real comparison needs the real data, so run with `--rule max` and `--rule sum` and compare.
- **Prior volume:** the fractional Δε and the conditional ε_1 change the prior volume relative to uniform-in-absolute, so Bayes factors depend on those choices. `PRIOR["delta_mode"]` and `PRIOR["eps1_mode"]` switch them.
- **GW170817 term:** it takes the (m1, m2) pairs from the same posterior samples as the Λ KDE. That is an approximation.
- **Crust:** the crust is taken to be included in the hadronic table, as stated.